# UID EXPERIMENT 3

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**1.COMMAND LINE INTERFACE:**

**PYTHON CODE:**

```
import os import
sys

def rename_file(old_name, new_name):
try:
    os.rename(old_name, new_name)    print(f"File
renamed from '{old_name}' to '{new_name}'")    except
FileNotFoundError:
    print(f"Error: File '{old_name}' not found.")
except PermissionError:
    print("Error: Permission denied.")
except Exception as e:
    print(f"An unexpected error occurred: {e}")

def main():    if
len(sys.argv) != 3:
    print("Usage: python rename_file_cli.py <old_filename> <new_filename>")
sys.exit(1)

  old_name = sys.argv[1]
```

```
    new_name = sys.argv[2]


    rename_file(old_name, new_name)


if __name__ == "__main__":
    main()
```

**OUTPUT:**

```
PS C:\Users\hi\OneDrive\Desktop\UID Experiments> python rename_file_cli.py main.txt sample.txt
File renamed from 'main.txt' to 'sample.txt'
```

## 2.GRAPHICAL USER INTERFACE:

**PYTHON CODE:**

```
import tkinter as tk from tkinter
import messagebox import os


# Function to rename the file def
rename_file():
    old_name = old_filename_entry.get().strip()
new_name = new_filename_entry.get().strip()


    # Check if input fields are empty
if not old_name or not new_name:
        messagebox.showwarning("Warning", "Please enter both filenames!")
return
```

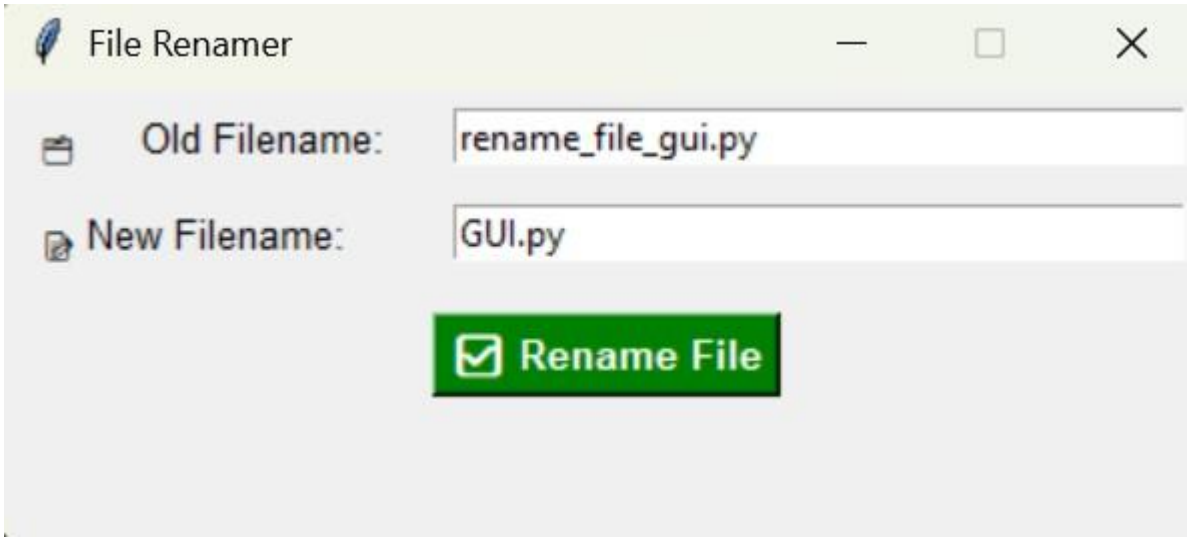
```
    # Check if the old file exists     if
not os.path.exists(old_name):
        messagebox.showerror("Error", f"File '{old_name}' not found.")
return

    # Check if the new file already exists
if os.path.exists(new_name):
        overwrite = messagebox.askyesno(
            "Warning", f"'{new_name}' already exists. Overwrite?"
        )
        if not overwrite:
            return

try:
        os.rename(old_name, new_name)
messagebox.showinfo(
            "Success", f"File renamed from '{old_name}' to '{new_name}'"
        )
    except Exception as e:
        messagebox.showerror("Error", f"An error occurred: {e}")

# Create main window
root = tk.Tk() root.title("File
Renamer")
root.geometry("400x150")
```

```
root.resizable(False,   False)
# Fixed window size


# Labels tk.Label(root, text="Old Filename:", font=("Arial",
10)).grid(     row=0, column=0, padx=10, pady=5, sticky="w"
)
tk.Label(root, text="New Filename:", font=("Arial", 10)).grid(
row=1, column=0, padx=10, pady=5, sticky="w"
)

# Entry fields old_filename_entry = tk.Entry(root, width=40)
old_filename_entry.grid(row=0, column=1, padx=10, pady=5)

new_filename_entry = tk.Entry(root, width=40) new_filename_entry.grid(row=1,
column=1, padx=10, pady=5)

# Styled Button rename_button
= tk.Button(
    root,
    text="Rename File",
bg="green",     fg="white",
font=("Arial", 10, "bold"),
command=rename_file
)
rename_button.grid(row=2, column=0, columnspan=2, pady=10)
```

# Run the GUI event loop root.mainloop()

**OUTPUT:**

```
PS C:\Users\hi\OneDrive\Desktop\UID Experiments> python rename_file_gui.py
```




**3.VOICE USER INTERFACE:**

**PYTHON CODE:**

```
import speech_recognition as sr import
os

def rename_file_from_voice_command(command):     try:
        command = command.lower()

        if "rename" not in command or "to" not in command:
            print("Invalid command. Say: 'Rename oldfile.txt to newfile.txt'")
return
```

```
        # Extract filenames using split        before_to,
after_to = command.split("to", 1)        old_name =
before_to.replace("rename", "").strip()        new_name
= after_to.strip()

        # Check if old file exists        if
not os.path.exists(old_name):
            print(f"Error: File '{old_name}' not found.")
return

        # Check if new file already exists
if os.path.exists(new_name):
            print(f" Error: '{new_name}' already exists.")
return

        os.rename(old_name, new_name)        print(f" File
renamed from '{old_name}' to '{new_name}'")

    except Exception as e:
print(f"Error: {e}")

def listen_for_command():
    recognizer = sr.Recognizer()

    try:        with sr.Microphone() as
source:
```

```
        print("Listening for command to rename a file...")
recognizer.adjust_for_ambient_noise(source)          audio =
recognizer.listen(source)

    command = recognizer.recognize_google(audio)
print(f" Command received: {command}")
rename_file_from_voice_command(command)

  except sr.UnknownValueError:
    print("Sorry, I couldn't understand the command.")    except
sr.RequestError as e:
    print(f"Speech recognition service error: {e}")
except OSError:
    print(" Microphone not found or access denied.")

if __name__ == "__main__":
listen_for_command()
```

**OUTPUT:**

```
PS C:\Users\hi\OneDrive\Desktop\UID Experiments> python rename_file_vui.py
🎤 Listening for command to rename a file...
🎙Command received: rename file
❌ Invalid command. Say: 'Rename oldfile.txt to newfile.txt'
```

```
Requirement already satisfied: pyaudio in .\vui_env\Lib\site-packages (0.2.14)
PS C:\Users\hi\OneDrive\Desktop\UID Experiments> python rename_file_vui.py
🎤 Listening for command to rename a file...
❌ Sorry, I couldn't understand the command.
```

```
PS C:\Users\hi\OneDrive\Desktop\UID Experiments> python rename_file_vui.py
🎤 Listening for command to rename a file...
🎙Command received: rename sample.txt to main dot text
✅ File renamed from 'sample.txt' to 'main dot text'
```

## 4.USER SATISFACTION COMPARISON:

## PYTHON CODE:

```
def survey():
    print("Rate your satisfaction with the following interfaces (1-5):")


    try:
        # Get user input for each interface
cli_satisfaction = int(input("CLI (1-5): "))
gui_satisfaction = int(input("GUI (1-5): "))
vui_satisfaction = int(input("VUI (1-5): "))


        # Ensure valid ratings        if not (1
<= cli_satisfaction <= 5 and
                1 <= gui_satisfaction <= 5 and
1 <= vui_satisfaction <= 5):
print("Please enter ratings between 1
and 5 only.")            return


        # Display the ratings
print("\nYour satisfaction ratings:")
print(f"CLI: {cli_satisfaction}")
print(f"GUI: {gui_satisfaction}")
print(f"VUI: {vui_satisfaction}")
```

```
        # Calculate the average satisfaction
avg_satisfaction = (            cli_satisfaction + gui_satisfaction
+ vui_satisfaction
        ) / 3
        print(f"\nAverage Satisfaction Score: {avg_satisfaction:.2f}")


    except ValueError:
        print("Invalid input! Please enter numbers between 1 and 5.")


# Run the survey function if
__name__ == "__main__":
    survey()
```

**OUTPUT:**

```
PS C:\Users\hi\OneDrive\Desktop\UID Experiments> python survey.py
Rate your satisfaction with the following interfaces (1-5):
CLI (1-5): 4
GUI (1-5): 5
VUI (1-5): 2

Your satisfaction ratings:
CLI: 4
GUI: 5
VUI: 2

Average Satisfaction Score: 3.67
PS C:\Users\hi\OneDrive\Desktop\UID Experiments>
```